

Algebraic

Week 1C

Overview

Factorisation

- Key Ideas*
- *Simple Common Factors*
 - *Compound Common Factor*
 - *Difference of two Squares*
 - *Sum and Difference of two Cubes*

Introduction - Factorisation

Discussion Question:

What are the factors of 21?

$$7 \times 3 = 21 \quad 7 \text{ \& 3 are the factors}$$

Expand $5(x + 2)$.

$$= 5x + 10$$

What are the factors of $5x + 10$?

$$\begin{array}{c} \downarrow \quad \downarrow \\ 5 \times x + 5 \times 2 = 5(x + 2) \end{array}$$

Finding factors, or *factorisation*, of algebraic expressions is important for solving equations, adding, subtracting, and simplifying fractions and calculus. There are a few techniques that can be learnt, but practice is the best way to master this topic.

Factorising *is the opposite of expanding*.

It is the *reverse of using the distributive law*. The distributive law says that

$$a(b + c) = ab + ac$$

When we factorise an expression, we go the other way: we start with an expression like

$$ab + ac$$

and rewrite it as

$$a(b + c).$$

To do this, we look for a *common factor in all terms in the expression* and then put it out the front and the remaining factor from each term inside the brackets – this is the using the distributive in reverse.

In the above expression

$$ab + ac,$$

there is a common factor of a , putting the common factor of a out the front and the remaining factor from each term inside the brackets

$$a(b + c).$$

We have *re-written the expression as a product* using the distributive law. After the common factor has been taken out, the terms remaining in the brackets should have no other factors in common.

Note: Since factorisation is the opposite of expanding, you can always check whether you have factorized correctly by expanding the factors, that is do the opposite, to make sure that you end up with the original algebraic expression.

Key Idea - Simple Common Factors

Factorising simple common factors consists of finding the ***highest common factor (HCF) of all terms in the expression*** and then put it out the front and the remaining factor from each term inside the brackets - then apply ***distributivity in reverse***.

If a , b and c are real numbers, then

$$ab + ac = a(b + c)$$

Note that the ***HCF may be a term in brackets as well***, such as $(x + y)$, rather than just a single number or variable.

Examples: Factorise the following:

$$3m + 3n = 3(m + n)$$

$$9a - 3a^2b + 3ab = 3a(3 - ab + b)$$

$$8a^2x^5 + 12a^2x^3 = 4a^2x^3(2x^2 + 3)$$

$$(a + b)^3 + x(a + b) = (a + b)[(a + b)^2 + x]$$

Exercises: Fully factorise the following expressions:

(a) $2x^2 + 4x = 2x(x + 2)$

(b) $4x^2yz + 8xyz^2 - 2xy^2z = 2xyz(2x + 4z - y)$

(c) $-a^2 - ab - 5c = -(a^2 + ab + 5c)$

Exercises: Fully factorise the following expressions:

$$(a) \quad 7(\underline{x-2y}) + 3x(\underline{x-2y}) = (x-2y)(7+3x)$$

$$(b) \quad 3x(a+b)^4 - 9x^2(a+b)^2 = 3x(a+b)^2 [(a+b)^2 - 3x]$$

Key Idea - Compound Common Factor

As we know, factorising uses the distributivity law in reverse. We can use this same idea to *factorise more complicated expressions*. It can be harder to spot the factors, so it is helpful to do it in steps, at least until you are able to recognise the pattern.

Examples: Factorise the following:

$$\begin{aligned}ab + ac + 2b + 2c &= a(b + c) + 2(b + c) \\ &= (a + 2)(b + c)\end{aligned}$$

$$\begin{aligned}3x + 3y + ax + ay &= 3(x + y) + a(x + y) \\ &= (3 + a)(x + y)\end{aligned}$$

$$\begin{aligned}ab + 3bc - 4a + 12c &= b(a + 3c) + 4(a + 3c) \\ &= (b + 4)(a + 3c)\end{aligned}$$

Exercises: Fully factorise the following expressions:

$$\begin{aligned} \text{(a)} \quad x^3 - 3x^2 + 2x - 6 &= x^2(x-3) + 2(x-3) \\ &= (x-3)(x^2 + 2) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad ab + 3bc - 4a + 12c &= b(a+3c) + 4(a+3c) \\ &= (a+3c)(b+4) \end{aligned}$$

Key Idea - Difference of Two Square

The *difference of two squares* is a special algebraic identity that states:

$$a^2 - b^2 = (a - b)(a + b)$$

It applies when you have two perfect squares separated by a minus sign.

The handy trick to remember is that

$$\begin{aligned}(a - b)(a + b) &= a(a + b) - b(a + b) \\ &= a^2 + ab - ba - b^2 \\ &= a^2 - b^2\end{aligned}$$

So, you can factorise a difference of two squares at a glance by using the above identity backwards:

Discussion Question:

Expand $(x-3)(x+3)$?

$$\begin{aligned} &= x(x+3) - 3(x+3) \\ &= x^2 + 3x - 3x - 9 \\ &= x^2 - 9 \end{aligned}$$

Now factories $x^2 - 3^2$?

$$\rightarrow \therefore (x-3)(x+3)$$

For example, Factorise the following:

$$x^2 - 4 = x^2 - 2^2$$

$$= (x-2)(x+2)$$

$$(2y)^2 - (y+1)^2 = (2y - (y+1))(2y + (y+1))$$

$$= (2y - y - 1)(2y + y + 1)$$

$$= (y-1)(3y+1)$$

Exercises: Factorise fully:

$$\begin{aligned} \text{(a)} \quad a^2 - 25 &= a^2 - 5^2 \\ &= (a-5)(a+5) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad 9y^2 - 16 &= (3y)^2 - 4^2 \\ &= (3y-4)(3y+4) \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad (x+2y)^2 - 16x^2 &= (x+2y)^2 - (4x)^2 \\ &= (x+2y-4x)(x+2y+4x) \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad x^4 - y^4 &= (x^2)^2 - (y^2)^2 \\ &= (x^2 - y^2)(x^2 + y^2) \end{aligned}$$

$$= (x^2 - y^2)(x^2 + y^2) = (x-y)(x+y)(x^2 + y^2)$$

Discussion Question:

Could you use this same technique to factorise $x^2 + 16$?

$x^2 + 4^2$ can not
be
factorised

Note: $x^2 + y^2$ cannot be factorised!!!!

Key Idea - Sum and Difference of Two Cubes

Discussion Question:

Expand $(x + 4)(x^2 - 4x + 16)$? What do you notice?

$$\begin{aligned} & x(x^2 - 4x + 16) + 4(x^2 - 4x + 16) \\ &= x^3 - \cancel{4x^2} + \cancel{16x} + \cancel{4x^2} - \cancel{16x} + 64 \\ &= x^3 + 64 \end{aligned}$$

To factorise the *sum or difference between two cubes*, the following formulae are used:

$$a^3 + b^3 = (a + b) (a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Note: To use them, identify the terms being cubed ($a - b$), then substitute them into the correct formula. A helpful mnemonic is "SOAP" (Same, Opposite, Always Positive) for remembering the signs in the factors.

Example,

Factorise the following:

$$\begin{aligned}x^3 - 27 &= x^3 - 3^3 \\ &= (x - 3)(x^2 + 3x + 9)\end{aligned}$$

Exercises: Factorise fully:

$$\begin{aligned} \text{(a) } 1 + 27x^3 &= 1^3 + (3x)^3 \\ &= (1 + 3x)(1^2 - 1 \times 3x + (3x)^2) \\ &= (1 + 3x)(1 - 3x + 9x^2) \end{aligned}$$

$$\begin{aligned} \text{(b) } 3(x^3 - 8) &= 3(x^3 - 2^3) \\ &= 3(x - 2)(x^2 + 2x + 4) \end{aligned}$$